



DEEP LEARNING

CSA-3007

What is Learning

❑ What is learning ?

- The learning is the process of examining the change that occurs in the surrounding environment and based on historical data establishing some rule to predict the future.

❑ What is machine learning ?

- Machine learning is a growing technology which enables computers to learn automatically from past data. Machine learning uses various algorithms for building mathematical models and making predictions using historical data or information.

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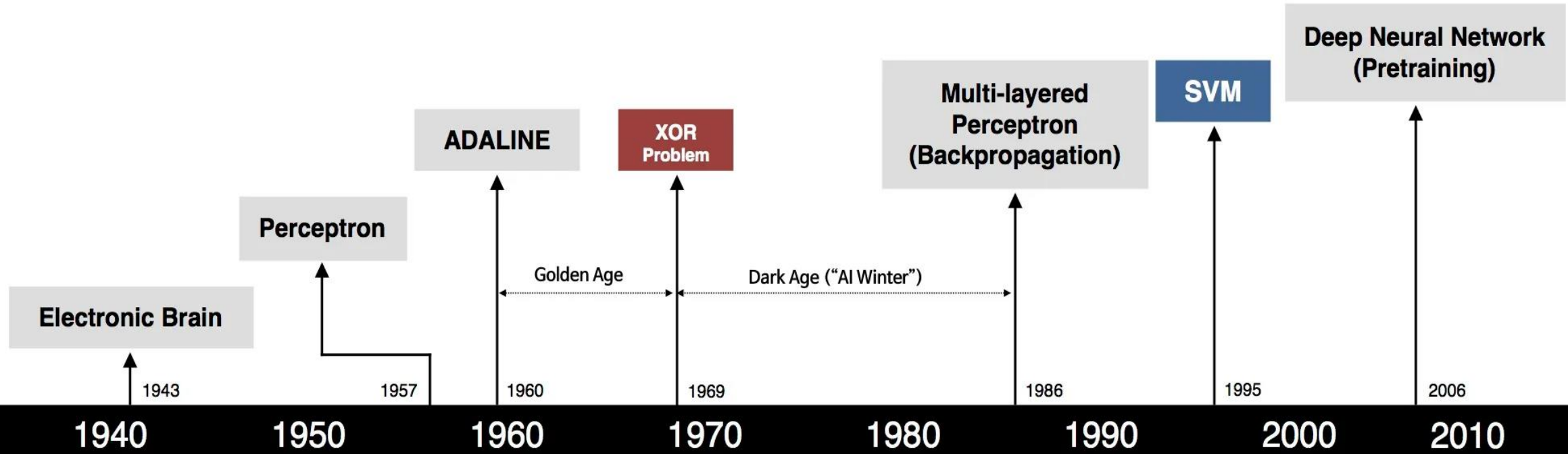
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What is Learning

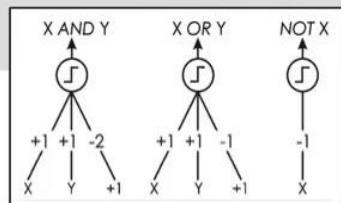
❑ What is Deep learning ?

- Deep Learning is a subfield of Machine Learning that focuses on learning hierarchical representations of data using multi-layer neural networks.

Historical Trends in Deep Learning



S. McCulloch – W. Pitts



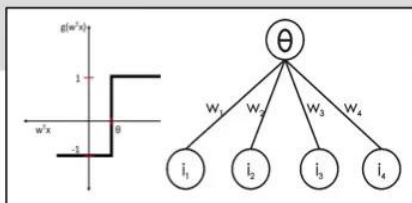
- Adjustable Weights
- Weights are not Learned



F. Rosenblatt



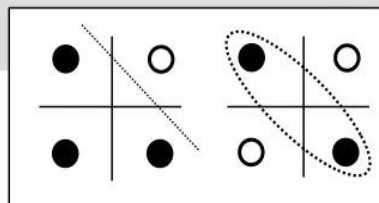
B. Widrow – M. Hoff



- Learnable Weights and Threshold



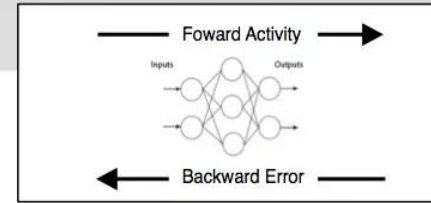
M. Minsky – S. Papert



- XOR Problem



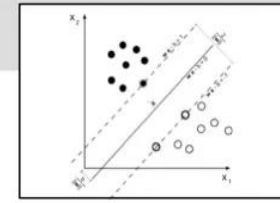
D. Rumelhart – G. Hinton – R. Williams



- Solution to nonlinearly separable problems
- Big computation, local optima and overfitting



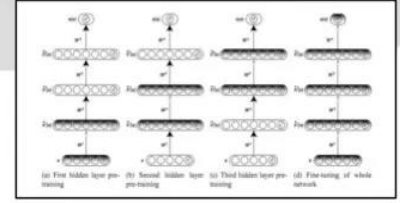
V. Vapnik – C. Cortes



- Limitations of learning prior knowledge
- Kernel function: Human Intervention



G. Hinton – S. Ruslan



- Hierarchical feature Learning

Historical Trends in Deep Learning

Evolution Overview

- 1950s–1960s: Foundations
 - Artificial neurons (Perceptron)
 - Rule-based AI systems
 - Limited computation and data
- 1970s–1980s: First AI Winter
 - Neural networks struggled with complex problems
 - Lack of training algorithms and processing power
- 1986: Backpropagation
 - Efficient training of multi-layer networks
 - Renewed interest in neural networks

Historical Trends in Deep Learning

- 1990s–2000s: Machine Learning Era
 - Support Vector Machines
 - Decision Trees
 - Shallow learning models
- 2010–Present: Deep Learning Revolution
 - Big Data availability
 - GPUs and parallel computing
 - Breakthroughs in vision, speech, NLP.

Linear Algebra for Deep Learning

- Linear algebra forms the mathematical foundation of deep learning models.

□ Scalars

- A scalar is a single numerical value.
- Denoted by lowercase letters.

Example:

- Learning rate ($\eta = 0.01$)
- Bias value ($b=2.5$)

Used in:

- Bias terms
- Hyperparameters

Linear Algebra for Deep Learning

- Vectors
- A vector is an ordered list of numbers.
- Used to represent:
 - Input features
 - Weights of neurons.

$$x=[x_1,x_2,x_3,\dots,x_n]$$

- Applications:
 - Feature representation
 - Gradient vectors

Linear Algebra for Deep Learning

- **Matrices**

- A matrix is a 2D array of numbers.
- Used to represent:
 - Datasets
 - Weight connections between layers.

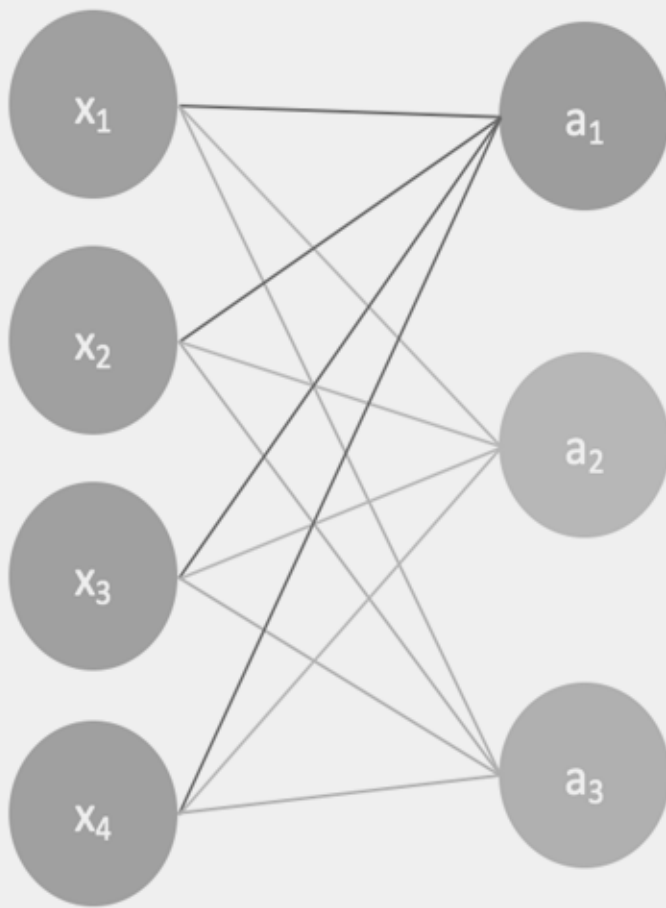
$$W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \\ w_{31} & w_{32} \end{bmatrix}$$

- Applications::
 - Weight matrices in neural networks
 - Batch input representation

- Matrix Operations in DL:
 - Matrix multiplication
 - Transpose
 - Inverse (limited usage)

Input layer

Output layer



A simple neural network

$$\begin{bmatrix} w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} b \\ b \\ b \end{bmatrix} = \begin{bmatrix} w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \\ w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \\ w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \end{bmatrix} \xrightarrow{\text{activation}} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

Linear Algebra for Deep Learning

- **Tensors**
- A tensor is a generalization of scalars, vectors, and matrices.
- Used extensively in deep learning frameworks (TensorFlow, PyTorch).
- Used for multi-dimensional data.

Order	Object
0	Scalar
1	Vector
2	Matrix
≥ 3	Tensor

- Example: RGB Image $\rightarrow$ 3D tensor (height $\times$ width $\times$ channels).

Linear Algebra for Deep Learning

- Norms
- Sometimes we need to measure the size of a vector.
- In machine learning, we usually measure the size of vectors using a function called a norm.
- A norm measures the size or magnitude of a vector or matrix.

$$\|x\|_p = (\sum |x_i|^p)^{\frac{1}{p}}$$

Linear Algebra for Deep Learning

- Norms

- Common Norms:

- L1 Norm:

$$\|x\|_1 = \sum |x_i|^1$$

Encourages sparsity

- L2 Norm

$$\|x\|_2 = \sqrt{\sum x_i^2}$$

Most commonly used

Applications:

Regularization

Avoiding overfitting



What is Learning

Neural Networks

Structure of a Neuron

Each artificial neuron performs:

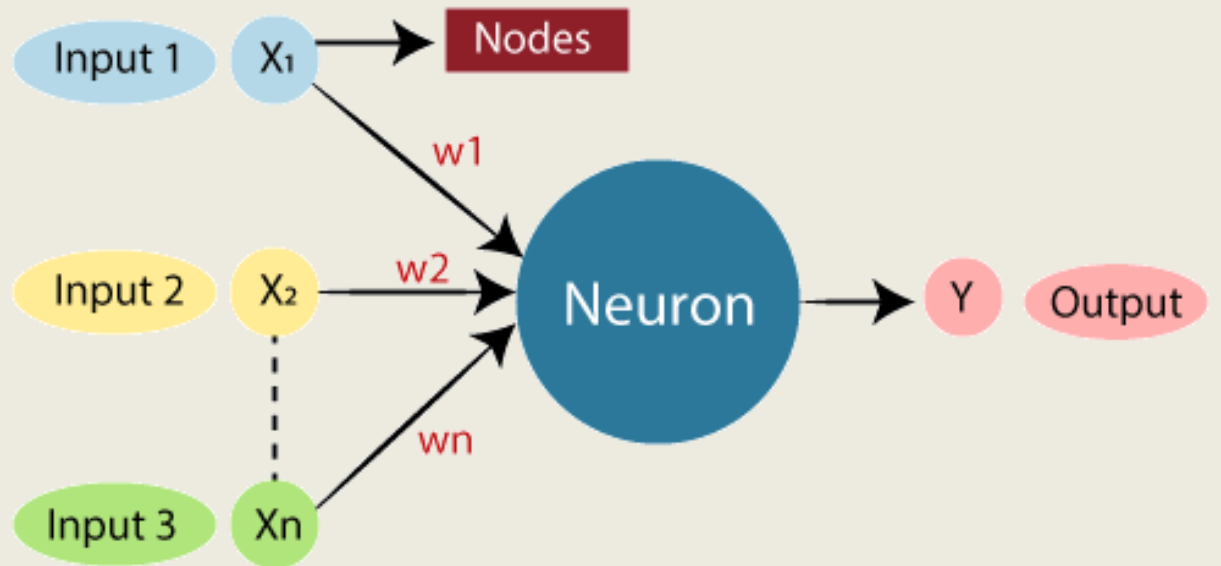
1. **Input Collection** (features from data)

2. **Weight Multiplication**

3. **Summation** ($\sum w_i x_i + b$)

4. **Activation Function**

5. **Output Generation**





Neural Networks

